Lohit Gali

Software Development Engineer in Test

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PROFESSIONAL SUMMARY

Software Development Engineer in Test with 5+ years of programming experience in Python/C/C++, specializing in the technical design and implementation of automated tests. Expertise in applying AI technologies for test plans creation, test cases development and test cases automation, with a strong background in developing Python scripts, testing RESTful APIs, and debugging complex systems to identify defects early in the software development lifecycle.

TECHNICAL SKILLS

Testing & Automation: Automated Tests, End-to-End Testing, Performance Testing, Load Testing, Compatibility Testing, Test Case Design, Defect Reporting, Code Coverage Analysis, CI/CD Integration, Debugging Tools (WinDBG/gdb)

Programming Languages: Python, C, C++, SQL, Bash

AI & Machine Learning: AI Technologies for Testing, ML/DL based applications (project experience), Prompt Engineering, LLM Integration (RAG, Agents)

Tools & Platforms: AWS, Azure (ADLS Gen2, Fabric, Logic Apps), Docker, Git, GitHub Actions, Azure DevOps, Jenkins, Elasticsearch, N8n, Power BI, Power Automate, Dynamics 365

Systems & Hardware: OS Fundamentals, PC Hardware and Troubleshooting, GPU Domain, GPUs Devices and Technologies (DLSS, Frame Generation, Reflex, CUDA, G-Sync), Object-Oriented Programming Concepts

PROFESSIONAL EXPERIENCE

Senior SDET & Automation Engineer | Epik Solutions

Jul 2025 – Present

• Led the technical design and implementation of automated tests for critical financial data pipelines, ensuring data integrity and adherence to regulatory standards by identifying defects early in the software development lifecycle, improving data quality by 90% for internal reporting.

• Developed comprehensive automated end to end tests using Python and C++ for financial reporting systems, reducing manual testing effort by 70% and accelerating release cycles for key stakeholders.

• Engineered robust test frameworks incorporating AI technologies for test plans creation and test cases automation, leveraging LLM agents to predict potential failure points and enhance test coverage for complex financial models.

• Built specialized tools using Python scripts and Object-Oriented Programming concepts to accelerate execution workflows for asset reporting, reducing test setup time by 45% and providing data driven insights to development teams.

• Executed performance and load testing on data ingestion services to ensure applications could handle high traffic and usage, proactively identifying bottlenecks and optimizing system scalability to support 1M+ daily transactions.

• Collaborated with multiple development teams to define test strategies and improve test code coverage across NVIDIA's device driver software and SDKs (simulated environment), resulting in a 25% reduction in post-release defects.

• Utilized debugging tools like WinDBG/gdb to identify root causes of critical defects in data processing logic, providing detailed reports and resolutions that improved system reliability and maintainability for production environments.

Senior SDET, Cloud & Compliance | Samay Systems LLC

Jul 2024 – Jun 2025

• Designed and implemented automated tests for a SaaS compliance platform, focusing on cloud security and regulatory adherence, detecting critical vulnerabilities prior to deployment and reducing compliance audit effort by 30%.

• Developed Python scripts and applications that brought data driven insights to compliance workflows, automating evidence collection from AWS, Azure, and GitHub via secure APIs, which accelerated audit preparation by 60%.

• Applied AI technologies for test cases development, leveraging LLMs to generate realistic compliance scenarios and validate system responses, enhancing the precision of automated tests by 40% against complex regulatory requirements.

• Performed testing of RESTful APIs for compliance data services, conducting performance and load testing to ensure applications could handle high traffic and usage, maintaining SLA compliance for 99.9% uptime for enterprise clients.

• Implemented robust debugging strategies using OS fundamentals and problem solving skills to quickly isolate and resolve issues within cloud-based compliance services, reducing critical bug resolution time by 50% and improving system reliability.

SDET & Enterprise Automation | Arvam Tech

Jul 2023 – May 2024

• Engineered and maintained automated end to end tests for an Enterprise Asset Management System, utilizing Python and SQL to validate data consistency and workflow accuracy, resulting in a 20% reduction in manual regression testing time.

• Developed utility scripts and tools using Python/C++ to automate and optimize the testing workflows for asset tracking and operational datasets, leading to a 30% improvement in testing efficiency.

• Implemented data validation and reconciliation tests for asset inventory data, ensuring high fidelity for analytics-ready formats used in trend analysis, reducing reporting discrepancies by 15%.

• Designed and integrated automated compatibility tests across various interfaces of the asset management system, verifying seamless operation between Power BI dashboards and Dynamics 365, improving system interoperability.

• Participated in peer code reviews, providing feedback on performance, scalability, and correctness of test automation frameworks, enhancing the maintainability and reliability of the codebase for the development team.

Associate SDET & Solutions | RHYM Tech

Aug 2022 – May 2023

• Developed analytical models using Python and SQL to support automated tests for a Cybersecurity Threat Intelligence Platform, validating high-volume telemetry and event data against predefined security policies.

• Executed manual and automated tests for data ingestion pipelines, identifying and reporting defects early to ensure the accuracy and completeness of threat intelligence data, improving data quality by 25%.

• Applied Object-Oriented Programming concepts in developing Python scripts for data cleaning, transformation, and feature extraction, creating test data sets that supported robust downstream ML tasks and anomaly detection.

• Assisted in the technical design and implementation of tests for new security features, ensuring maintainable, reliable, and well detailed code that contributed to a 10% reduction in integration defects.

Associate Software Developer in Test | ZeMoSo Technologies

Jul 2021 – Aug 2022

• Developed automated tests for modern web application features using JavaScript and Python, ensuring responsive design and cross-browser compatibility, reducing UI-related defects by 15%.

• Implemented backend API tests using Python, verifying data access layers and ensuring correctness and performance of UI-driven workflows for a Biotech LabOps Automation platform.

• Supported data science initiatives by building automated data preparation and exploratory analysis scripts using Python, ensuring data quality for basic modeling tasks and analytical pipelines.

• Utilized problem solving skills to debug issues and validate application behavior during production deployments, monitoring early usage signals and addressing issues discovered after release to maintain system stability.

EDUCATION

Master’s in Computer Science | Oklahoma State University, Stillwater, OK

Bachelor’s in Computer Science | JNTU Hyderabad, India